

Question:
Bivalent meningococcal vaccine is ?
Answer Choices:
A. A Y
B. A C
C. C y
D. A W-135

Reasoning + Answer:

<think>
The bivalent meningococcal vaccine is a type of vaccine that protects against two serogroups of Neisseria meningitidis. The vaccine typically includes components that target serogroups A and W, which are the most common causes of meningococcal disease. The bivalent vaccine does not include serogroup C, as that is typically covered by a monovalent vaccine. The "Y" and "W-135" components are also included in some meningococcal conjugate vaccines, but they are not the primary components of the bivalent vaccine.

Considering the options provided, the correct answer would be the one that includes serogroups A and W, which are the primary components of the bivalent meningococcal vaccine.

Option A. A Y includes serogroups A and Y, which is not the correct composition of the bivalent meningococcal vaccine.

Option B. A C includes serogroups A and C, which is not the correct composition of the bivalent meningococcal vaccine.

Option C. C Y includes serogroups C and Y, which is not the correct composition of the bivalent meningococcal vaccine.

Option D. A W-135 includes serogroups A and W-135, which is the correct composition of the bivalent meningococcal vaccine.

Based on the information provided, the correct answer is the one that includes serogroups A and W-135, which are the primary components of the bivalent meningococcal vaccine.

</think>
<answer>
D. A W-135
</answer>